Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses partial fractions with linear denominators $\dfrac{6x+1}{6x^2-7x+2}=\dfrac{A}{ax+b}+\dfrac{B}{cx+d}$	AO3.1a	M1	$\dfrac{6x+1}{6x^2-7x+2}=\dfrac{A}{3x+2}+\dfrac{B}{2x-1}$ $A(2x-1)+B(3x-2)=6x+1$
Obtains correct linear denominators	AO1.1b	B1	$x=\dfrac{2}{3},\ A\left(\dfrac{1}{3}\right)=5$ so $A=15$
Obtains at least one numerator correct (using any valid method, eg equating coefficients or substitution of values)	AO1.1b	A1	$x=\dfrac{1}{2},\ B\left(-\dfrac{1}{2}\right)=4$ so $B=-8$ $\int_1^2 \dfrac{15}{3x-2}-\dfrac{8}{2x-1}\,dx$ $=\left[5\ln(3x-2)-4\ln(2x-1)\right]_1^2$
Obtains partial fractions completely correct	AO1.1b	A1	$=5\ln(4)-4\ln(3)-(5\ln(1)-4\ln(1))$ $=10\ln(2)-4\ln(3)$
Integrates 'their' partial fractions, must include logs $p\ln(ax+b)+q\ln(cx+d)$	AO1.1a	M1	
'Their' integral correct (ignore limits)	AO1.1b	A1F	
Substitutes limits into 'their' integral	AO1.1a	M1	
Correct final answer in correct form **CAO**	AO1.1b	A1	
			Total 8 marks

Q2.

(a) (i) $5x - 6 = A(x - 3) + Bx$

Multiply by denominator and use two values of x.

M1

$x = 0 \quad x = 3$

$A = 2 \quad B = 3$

A1

Alternative: equate coefficients

$-6 = -3A \quad 5 = A + B$

Set up and solve simultaneous equations for values of A and B.

(M1)

$A = 2 \quad B = 3$

(A1)

2

(ii) $\left(\int \frac{2}{x} + \frac{3}{x-3}\,dx = \right) 2\ln x$

their A ln x

B1ft

$+ 3\ln(x - 3)(+C)$

their B ln $(x - 3)$ and no other terms; condone B ln $x - 3$

B1ft

2

(b) (i)

$$\begin{array}{r|l} & 2x^2 - x + 3 \\ 2x+1 & 4x^3 + 5x - 2 \\ & 4x^3 + 2x^2 \\ & -2x^2 + 5x \\ & -2x^2 - x \\ & 6x - 2 \\ & 6x + 3 \\ & -5 \end{array}$$

Division as far as $2x^2 + px + q$ with $p \neq 0$, $q \neq 0$, PI

M1

$p = -1$

PI by $2x^2 - x + q$ seen

A1

$q = 3$

PI by $2x^2 - x + 3$ seen

A1

$r = -5$

and must state $p = -1$, $q = 3$, $r = -5$ explicitly or write out full correct RHS expression

A1

Alternative 1:

$4x^3 + 5x - 2 = 4x^3 + (2 + 2p)x^2 + (p + 2q)x + q + r$

$2 + 2p = 0$

Clear attempt to equate coefficients, PI by $p = -1$

(M1)

$p + 2q = 5$

$q + r = -2$

$p = -1$

(A1)

$q = 3 \quad r = -5$

(A1A1)

Alternative 2:

$4x^3 + 5x - 2 = (2x + 1)(2x^2 + px + q) + r$

$x = -\frac{1}{2} \quad 4\times\left(-\frac{1}{2}\right)^3 + 5\left(-\frac{1}{2}\right) + 2 = r$

$x = -\frac{1}{2}$ *used to find a value for* r

(M1)

$r = -5$

(A1)

$p = -1,\ q = 3$

(A1A1)

4

(ii) $\left(\frac{4x^3 + 5x - 2}{2x + 1} =\right) 2x^2 + px + q + \frac{r}{2x+1}$

M1

$\frac{2}{3}x^3 - \frac{1}{2}x^2 + 3x + k\ln(2x + 1)\ (+ C)$

ft on p *and* q

A1ft

$\frac{2}{3}x^3 - \frac{1}{2}x^2 + 3x - \frac{5}{2}\ln(2x + 1)(+ C)$

CSO

A1

3

[11]

Q3.

(a) (i) $\int \frac{1}{3+2x} dx$

$= k \ln (3 + 2x)$

Where k is a rational number

M1

$= \frac{1}{2} \ln(3+2x) + c$

Or

if substitution $u = 3 + 2x$, $du = 2dx$

$\int = \int \frac{1}{u}\frac{du}{2} = k \ln u$

M1

$= \frac{1}{2} \ln(3+2x) + c$

A1

A1

2

(b) $u = x \quad dv = \sin\frac{x}{2}$

$\int \sin\frac{x}{2}(dx) = k\cos\frac{x}{2}, \frac{d}{dx}(x) = 1$

where k is a constant

M1

$du = 1 \quad v = -2\cos\frac{x}{2}$

All correct

A1

$\int = -2x\cos\frac{x}{2} - \int -2\cos\frac{x}{2}(dx)$

***Correct** substitution of their terms into parts formula (watch signs carefully)*

m1

$= -2x\cos\frac{x}{2} + 4\sin\frac{x}{2} + c$

CAO

A1

4

[6]

Q4.

(a) $1 = A(1 - x)^2 + B(1 - x)(3 - 2x) + C(3 - 2x)$

Attempt to clear fractions

M1

$$\left.\begin{array}{lll} x = 1 & x = \frac{3}{2} & x = 0 \\ C = 1 & 1 = A\left(-\frac{1}{2}\right)^2 & 1 = A + 3B + 3C \end{array}\right\}$$

Use any two (or three) values of x to set up two (or three) equations

m1

$A = 4$ $B = -2$ $C = 1$

Two values correct

A1

All values correct

A1

Alternative

$1 = A(1 - x)^2 + B(1 - x)(3 - 2x) + C(3 - 2x)$

(M1)

$1 = A + 3B + 3C$

$0 = -2A - 5B - 2C$

Set up three simultaneous equations

(m1)

$0 = A + 2B$

Two values correct

(A1)

$A = 4 \quad B = -2 \quad C = 1$

All values correct

(A1)

4

(b) $\int \frac{1}{2\sqrt{y}}\,\mathrm{d}y = \int \frac{4}{3-2x} - \frac{2}{1-x} + \frac{1}{(1-x)^2}\,\mathrm{d}x$

Separate using partial fractions; correct notation; condone missing integral signs but $\mathrm{d}y$ and $\mathrm{d}x$ must be in correct place.

ft on their A, B, C and on each integral.

B1ft

$\int \frac{1}{2\sqrt{y}}\,\mathrm{d}y = \sqrt{y} =$

OE $\int \frac{k}{\sqrt{y}}\,\mathrm{d}y = 2k\sqrt{y}$ *is B1*

B1

$-2\ln(3-2x)$

Condone missing brackets on one ln integral.

B1ft

$+ 2\ln(1-x)$

B1ft

$+ \frac{1}{1-x} (+C)$

Condone omission of +C

B1ft

$x = 0 \quad y = 0 \Rightarrow 0 = -2\ln 3 + 0 + 1 + C$

Use (0,0) to find C. Must get to C = …..

M1

$C = 2\ln 3 - 1$

Correct C found from correct equation.

C must be exact, in any form but not decimal.

A1

$$\sqrt{y} = 2\ln\left(\frac{3-3x}{3-2x}\right) + \frac{1}{1-x} - 1$$

Correct use of rules of logs to progress towards requested form of answer. C must be of the form r ln$s$ + t

m1

$$y^{\frac{1}{2}} = 2\ln\left(\frac{3-3x}{3-2x}\right) + \frac{x}{1-x}$$

OE
CSO condone B0 for separation

A1

9

[13]

Q5.

(a) $10x^2 + 8 = 2(x + 1)(5x - 1) +$

A and B terms correct

M1

$A(5x - 1) + B(x + 1)$

A1

$x = -1 \quad x = \frac{1}{5}$

$A = -3 \quad B = 7$

Use two values of x to find A and B, or set up and solve
$8 + 5A + B = 0$
$-2 - A + B = 8$

SC1 NWS A & B correct $\frac{4}{4}$

SC2 NWS A or B correct $\frac{1}{4}$

m1A1

4

(b) $$\int \frac{10x^2+8}{(x+1)(5x-1)}\,dx = \int 2 - \frac{3}{x+1} + \frac{7}{5x-1}\,dx$$

Use the partial fractions

M1

$= 2x + C$

B1

a In $(x + 1)$ + b In $(5x – 1)$
condone missing brackets

M1

$-3\ln(x+1)+\frac{7}{5}\ln(5x-1)$

F on A and B

A1F

4

[8]

Q6.

(a) (i) $7x – 3 = A(3x – 2) + B(x + 1)$

M1

$x = –1$ $x = \frac{2}{3}$

Substitute two values of x and solve for A and B

m1

$A = 2$ $B = 1$

Or solve $\left.\begin{array}{l} 7 = 3A + B \\ -3 = -2A + B \end{array}\right\}$ *condone one error*

A1

Alternative:
By cover up rule

$x = -1 \quad A = \frac{-7-3}{-5}$

$x = \frac{2}{3} \quad B = \frac{\frac{14}{3}-3}{\frac{5}{3}}$

$x = -1$ *and* $x = \frac{2}{3}$

and attempt to find A and B

(M1)

$A = 2 \qquad B = 1$

SC NMS A and B both correct 3/3

One of A or B correct 1/3

(A1, A1)

3

(ii) $\int \frac{7x-3}{(x+1)(3x-2)}\,dx =$

$p \ln (x + 1) + q \ln (3x - 2)$

Condone missing brackets

M1

$= 2 \ln (x + 1) + \frac{1}{3} \ln (3x - 2) (+ c)$

F on A and B; constant not required

A1F

2

(b) $\frac{6x^2 + x + 2}{2x^2 - x + 1} = \frac{6x^2 - 3x + 3 + 4x - 1}{2x^2 - x + 1}$

M1

$= 3 + \frac{4x-1}{2x^2 - x + 1}$

$P = 3$

B1

$Q = 4$ and $R = -1$

A1

Alternative:

$$\begin{array}{r} 3 \\ 2x^2 - x + 1 \overline{)\,6x^2 + x + 2} \\ \underline{6x^2 - 3x + 3} \\ 4x - 1 \end{array}$$

Complete division, with $ax + b$ remainder

(M1)

$P = 3$ stated

(B1)

$Q = 4$ and $R = -1$ stated or written as expression

(A1)

or

$6x^2 + x + 2 = P(2x^2 - x + 1) + Qx + R$
$= 2Px^2 + (Q - P)x + P + R$

Multiply across and equate coefficients or use numerical values of x

(M1)

$P = 3$

$P = 3$ stated

(B1)

$Q - P = 1$
$P + R = 2$
$Q = 4$ and $R = -1$

$Q = 4$ and $R = -1$ stated or written as expression

(A1) 3

[8]

Q7.

(a) (i) $\frac{2x+7}{x+2} = 2 + \frac{3}{x+2}$

B1B1 2

(ii) $\int \frac{2x+7}{x+2} = 3\ln(x+2) + 2x + C$

Either term correct

B1F

Both correct; constant required; condone missing bracket
ft on A, B

B1F 2

(b) (i) $28 + 4x^2$
$P(5 - x)^2 + Q(1 + 3x)(5 - x) + R(1 + 3x)$

M1

$x = 5$ $\qquad x = -\frac{1}{3}$

Two values of x used to find R and P.

m1

$R = 8$ $\qquad P = 1$

SC $R = 8$, $P = 1$ NMS can score B1, B1

A1

$x = 0 \Rightarrow 28 = 25P + 5Q + R$

Third value of x used to find Q

m1

$Q = -1$

A1

Alternative

$28 + 4x^2 =$

$P(5 - x)^2 + Q(1 - 3x)(5 - x) + R(1 + 3x)$

(M1)

$= (25P + 5Q + R) + (-10P + 14Q + 3R)x + (P - 3Q)x^2$

Collect terms and form equations

(m1)

$P - 3Q = 4$

$14Q + 3R - 10P = 0$

Correct equations

(A1)

$25P + 5Q + R = 28$

$P = 1 \quad Q = -1 \quad R = 8$

Solve for P Q and R

(m1)(A1)

5

(ii) $\displaystyle\int \frac{1}{1+3x} - \frac{1}{5-x} + \frac{8}{(5-x)^2}\,dx$

Use partial fractions

M1

$$= \frac{1}{3}\ln(1+3x) + \ln(5-x) + \frac{8}{5-x} + (C)$$

a ln (1 + $3x$) + b ln (5 – x)

m1

OE; both ln integrals correct; needs ( )

A1F

Other term correct
ft on their P, Q, R

A1F

4

SC: If no P, Q, R found in (b)(i), can gain method marks by inserting other values or retaining the letters (max 2/4)

[13]

Q8.

(a) (i) $y = 2x^2 - 8x + 3$

$$\left(\frac{dy}{dx} = \right) 4x - 8$$

B1

1

(ii)
$$\int_4^6 \frac{x-2}{2x^2 - 8x + 3}\,dx$$

$$= \frac{1}{4}\left[\ln\left|2x^2 - 8x + 3\right|\right]_4^6$$

M1 for k ln ($2x^2$ – $8x$ + 3); allow k ln u

M1A1

$$= \frac{1}{4}[\ln 27 - \ln 3]$$

Correct substitution into
k ln ($2x^2$ – $8x$ + 3) or 3, 27 into k ln u

m1

$$= \frac{1}{4}\ln 9$$

$$= \frac{1}{2}\ln 3$$

A1

4

(b) $\int x\sqrt{(3x-1)}\,dx$

$u = 3x - 1$ $\quad du = 3\,dx$

OE

B1

$\int = \left(\frac{1}{9}\right)\int\left(u^{\frac{3}{2}} + u^{\frac{1}{2}}\right)(du)$

$\int$2 terms in with rational indices

M1

$= \left(\frac{1}{9}\right)\left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{u^{\frac{3}{2}}}{\frac{3}{2}}(+c)\right]$

Must be 2 terms with correct indices

$\left(\textit{only ft for } x = \frac{u-1}{3}\right)$

A1F

$= \frac{2}{45}(3x-1)^{\frac{5}{2}} + \frac{2}{27}(3x-1)^{\frac{3}{2}} + c$

CSO OE

A1

4

[9]

Q9.

(a) $3 = k(3 + x + 3 - x)$

OE $\frac{A}{3-x} + \frac{B}{3+x} \Rightarrow 6A = 3 \quad 6B = 3$

M1

$k = \frac{1}{2}$

or eg put $x = 0, \frac{3}{9} = k\left(\frac{1}{3} + \frac{1}{3}\right) \Rightarrow k = \frac{1}{2}$

A1

2

(b) $\int_1^2 \frac{3}{9-x^2}dx = -\frac{1}{2}\ln(3-x) + \frac{1}{2}\ln(3+x)$

a ln (3 ± x)

M1

ft on k

A1F

$= \frac{1}{2}((\ln 5 - \ln 1) - (\ln 4 - \ln 2)) = \frac{1}{2}\ln\left(\frac{5}{2}\right)$

accept ln $\left(\frac{10}{4}\right)$

ft only for sign error in integral: $\frac{1}{2}\ln\left(\frac{5}{8}\right)$

A1F

3

[5]

Q10.

(a) $\frac{2}{(x^2-1)} = \frac{A}{x-1} + \frac{B}{x+1}$

$2 = A(x+1) + B(x-1)$

M1

$x = 1$ $\quad$ $x = -1$

use two values of x or equate coefficients and solve $A + B = 0$ and $A - B = 2$

m1

$A = 1$ $\quad$ $B = -1$

both A and B

A1

3

(b) $\int \frac{2}{x^2-1}dx = p\ln(x-1) + q\ln(x+1)$

ln integrals

M1

$= \ln(x - 1) - \ln(x + 1)$

F on A and B
condone missing brackets

A1F

2

(c) $$\int \frac{dy}{y} = \int \frac{2}{3(x^2 - 1)}\,dx$$

separate and attempt to integrate on one side

M1

$$\ln y = \frac{1}{3}\left(\ln(x - 1) - \ln(x + 1)\right) \quad (+C)$$

left hand side

A1

F from part (b) on right hand side

A1F

$(3, 1)$ $\quad \ln 1 = \frac{1}{3}\left(\ln 2 - \ln 4\right) + C$

use (3, 1) to attempt to find a constant

m1

$3 \ln y = \ln(x - 1) - \ln(x + 1) - (\ln 2 - \ln 4)$

$$3 \ln y = \left(\ln\left(\frac{x-1}{x+1}\right) + \ln 2\right)$$

$$\ln y^3 = \ln\left(\frac{2(x-1)}{x+1}\right)$$

$$y^3 = \frac{2(x-1)}{x+1}$$

CSO AG

A1

5

[10]

Q11.

(a) (i) $$\frac{3x-5}{x-3} = 3 + \frac{4}{x-3}$$

By division: B1 for 3, B1 for $\frac{4}{x-3}$ or $B = 4$
By partial fractions: M1 multiply by $x - 3$

and using 2 values of x, A1 both correct

B1, B1

2

(ii) $\int 3+\frac{4}{x-3}\,dx = 3x + 4\ln(x-3)(+c)$

M1 $\int 3+\frac{4}{x-3}\,dx$ *and attempt at integrals ft on A and B; condone omission of brackets around x – 3*

M1A1F

2

Alternative: By substitution $u = x - 3$

$\int \frac{3x-5}{x-3}\,dx = \int \frac{3u+4}{u}\,du$

Integral in terms of u

(M1)

= 3(x – 3) + 4ln(x – 3)

Correct, in x

(A1)

(b) (i) 6x – 5 = P(2x – 5) + Q(2x + 5)

Clear evidence of use of cover-up rule M2

M1

$x = \frac{5}{2}$ $x = -\frac{5}{2}$

m1

10 = 10Q – 20 = –10P

Q = 1 P = 2

A1

3

(ii) $\int \frac{2}{2x+5} + \frac{1}{2x-5}\,dx$

Attempt at ln integral (a ln (2x + 5) + b ln (2x – 5))

M1

$\ln(2x+5)+\frac{1}{2}\ln(2x-5)(+c)$

M1

ft on P and Q; must have brackets

A1F

3

[10]

Q12.

(a) (i) $f' = \frac{dy}{dx} = 4x^3 + 2$

B1

1

(ii) $\int \frac{2x^3+1}{x^4+2x}\,dx$

$= \frac{1}{2}\ln(x^4 + 2x)(+c)$

For k ln $(x^4 + 2x)$

M1

By substitution k ln u *M1*

correct *A1*

A1

2

(b) (i) $u = 2x + 1$

$du = 2\,dx$

B1

$\int x\sqrt{2x+1}\,dx =$

$\int\left(\frac{u-1}{2}\right)\sqrt{u}\,\frac{du}{2}$

Must be in terms of u only incl. du

M1

$$= \frac{1}{4}\int\left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right)\mathrm{d}u$$

AG

A1

3

(ii) $\int_0^4 \mathrm{d}x = \int_1^9 \mathrm{d}u$

Or changing u's to x's at end

B1

$$\frac{1}{4}\int u^{\frac{3}{2}} - u^{\frac{1}{2}} = \frac{1}{4}\left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} - \frac{u^{\frac{3}{2}}}{\frac{3}{2}}\right]$$

M1
A1

$$= \frac{1}{4}\left[\left(\frac{2}{5}(9)^{\frac{5}{2}} - \frac{2}{3}(9)^{\frac{3}{2}}\right) - \left(\frac{2}{5} - \frac{2}{3}\right)\right]$$

$= \frac{1}{4}$ [79.2 + $0.2\dot{6}$]

= 19.86

Sight of any of these 3 lines

= 19.9

AG

A1

4

[10]

Q13.

(a) $9x^2 - 6x + 5 = 3(3x - 1)(x - 1) + A(x - 1) + B(3x - 1)$

Or $3 + \dfrac{6x + 2}{(3x - 1)(x - 1)}$

B1

$x = 1 \qquad x = \frac{1}{3}$

Substitute $x = 1$ or $x = \frac{1}{3}$

M1

B = 4 A = −6

Or equivalent method (equating coefficients, simultaneous equations)

A1A1

4

(b) $\int = \int 3 - \frac{6}{3x-1} + \frac{4}{x-1}\,\mathrm{d}x$

Attempt to use partial fractions

M1

= $3x$ ...

B1

$-2\ln(3x-1) + 4\ln(x-1)(+c)$

p ln(3x − 1) + q ln(x − 1)
Condone missing brackets

M1

Follow through on A and B; brackets needed.

A1F

4

[8]